# Analyzing data on the level of cognitive processes - ESCoP 2025

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## Solutions: Exercise 3 - Set size effects

In this exercise we will implement the basic M3 model to the data sets from Oberauer (2019) "Working Memory Capacity Limits Memory for Bindings" in which participants completed a cued recall task. In two experiments, participant were asked to memorize lists. In different conditions, various numbers of words (set size 2, 4, 6, 8) were displayed within a list. At test, participants were cued with a list position and asked to select the word from a selection of words. This selection could contain apart from the correct word, other presented list words (IOP) or completely new words (NPL).

The data set for experiment 1 "exercise3_data_exp1.txt" and experiment 2 "exercise3_data_exp2.txt" are in the folder "data". Here is an overview what the individual variables in each data set refer to:

- id: participant ID
- setsize: setsize of words within a list (2, 4, 6 or 8) as a factor
- setsize_lin: setsize if words within a list (2, 4, 6 or 8) as a linear variable
- n_IIP: total number of correct responses in each trial
- n_IOP: total number of IOP responses in each trial
- n_NPL: total number of NPL in each trial
- IIP: number of correct responses (IIP)
- IOP: number of errors when a word from the list was reported, but at the incorrect position (IOP)
- NPL: number of errors when a word *not* from the list was reported (NPL)
- total: total number of IIP, IOP, and NPL

### Task 1: Experiment 1

We are interested in investigating how the set sizes affect participants memorizing the word lists in experiment 1 ("exercise3_data_exp1.txt"). Specifically, we are interested in examining how the cognitive processes, namely binding memory and item memory, change with increasing set sizes.

To this end, we will…

a) … examine and plot the proportions of correct responses in experiment 1 descriptively for the different set sizes. How do the proportions change for the different set sizes?

b) … implement the basic M3 model and plot the results (e.g., the model estimates, model predictions). How are binding memory and item memory affected by the different set sizes? Does this match your expectations?

#### Load libraries

```
# empty environment
rm(list = ls())

# load libraries
library("here")
library("bmm")
```

```r
library("dplyr")
library("tidyr")
library("brms")
library("ggplot2")
library("tidybayes")
library("readr")
```

### Load data

```r
# experiment 1
mydata_exp1 <- read.table(here("data", "exercise3_data_exp1.txt"))
mydata_exp1$setsize <- as.factor(mydata_exp1$setsize)
```

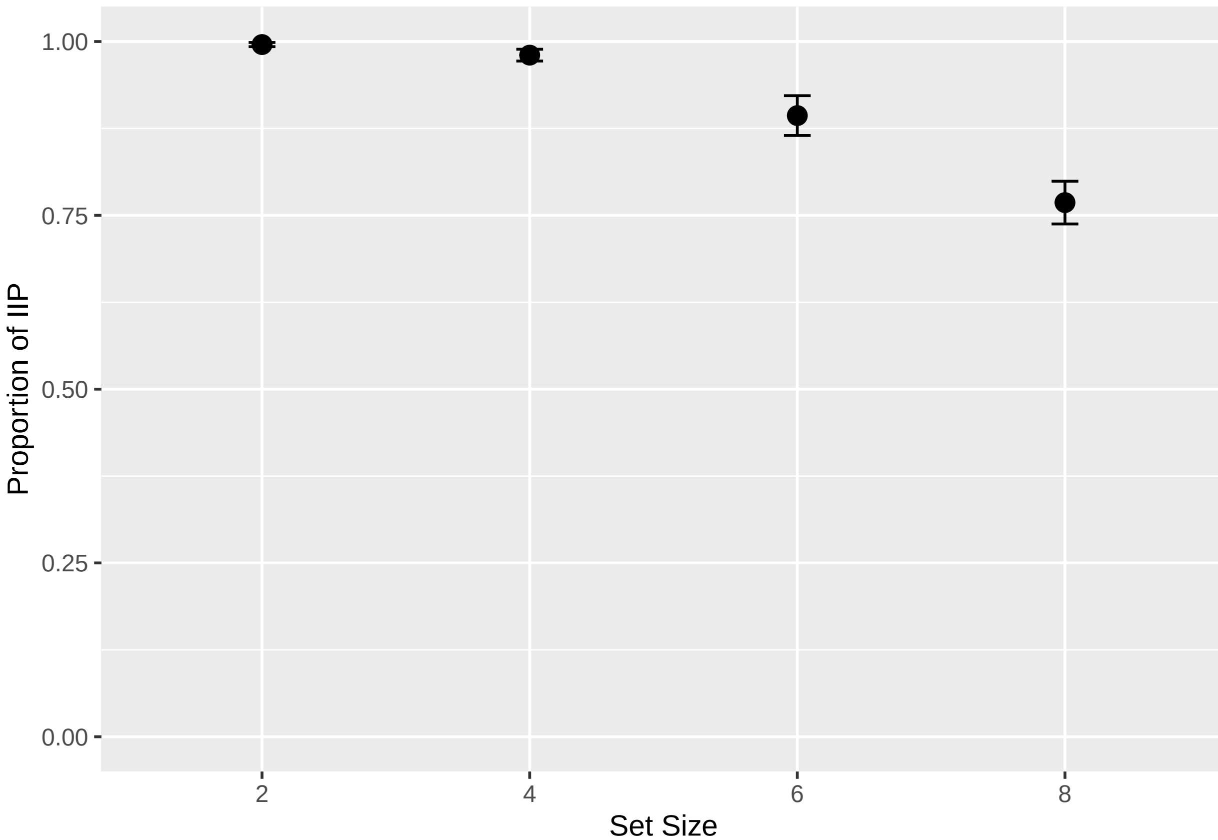
### Descriptive proportions

```r
# calculate proportions
prop_data <- mydata_exp1 %>%
  group_by(id, setsize) %>%
  summarise(prop = sum(IIP) / sum(total), .groups = "drop") %>%
  group_by(setsize) %>%
  summarise(
    mean_prop = mean(prop),
    se = sd(prop) / sqrt(n()),
    lower = mean_prop - 1.96 * se,
    upper = mean_prop + 1.96 * se,
    .groups = "drop"
  )

prop_data
```

```
## # A tibble: 4 x 5
##   setsize mean_prop      se lower upper
##   <fct>       <dbl>   <dbl> <dbl> <dbl>
## 1 2           0.996 0.00152 0.993 0.999
## 2 4           0.980 0.00432 0.972 0.989
## 3 6           0.893 0.0146  0.865 0.922
## 4 8           0.768 0.0157  0.738 0.799
```

```r
# plot proportions
ggplot(prop_data, aes(x = factor(setsize), y = mean_prop)) +
  geom_point(size = 3) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.1) +
  labs(y = "Proportion of IIP", x = "Set Size") +
  coord_cartesian(ylim = c(0, 1))
```




**Implement basic M3 model**

```
n_cores <- 4
n_iters <- 2000 # number of total iterations
n_warmup <- 1000 # number of warmups
n_chains <- 4 # number of chains
```

**Set model settings**

```
# model
m3_model_ss <- m3(resp_cats = c("IIP", "IOP", "NPL"),
               num_options = c("n_IIP", "n_IOP", "n_NPL"),
               choice_rule = "softmax",
               version = "ss")

# model formula
m3_formula_ss <- bmf(
  IIP ~ b + a + c,
  IOP ~ b + a,
  NPL ~ b,
  c ~ 1 + setsize + (1 + setsize | id),
  a ~ 1 + setsize + (1 + setsize | id)
)

# check priors
```

```r
default_prior(m3_formula_ss, data = mydata_exp1, model = m3_model_ss)
```

**Define model formula, priors and model**

```
##                  prior class      coef group resp dpar nlpar   lb   ub
##                 lkj(1)   cor
##                 lkj(1)   cor                id
##          normal(0,0.5)     b  setsize4                       a <NA> <NA>
##          normal(0,0.5)     b  setsize6                       a <NA> <NA>
##          normal(0,0.5)     b  setsize8                       a <NA> <NA>
##   student_t(3, 0, 2.5)    sd                                 a    0
##   student_t(3, 0, 2.5)    sd                id               a    0
##   student_t(3, 0, 2.5)    sd Intercept      id               a    0
##   student_t(3, 0, 2.5)    sd  setsize4      id               a    0
##   student_t(3, 0, 2.5)    sd  setsize6      id               a    0
##   student_t(3, 0, 2.5)    sd  setsize8      id               a    0
##                 (flat)     b                                 b
##            normal(0,2)     b  setsize4                       c <NA> <NA>
##            normal(0,2)     b  setsize6                       c <NA> <NA>
##            normal(0,2)     b  setsize8                       c <NA> <NA>
##   student_t(3, 0, 2.5)    sd                                 c    0
##   student_t(3, 0, 2.5)    sd                id               c    0
##   student_t(3, 0, 2.5)    sd Intercept      id               c    0
##   student_t(3, 0, 2.5)    sd  setsize4      id               c    0
##   student_t(3, 0, 2.5)    sd  setsize6      id               c    0
##   student_t(3, 0, 2.5)    sd  setsize8      id               c    0
##            normal(0,2)     b                                 c <NA> <NA>
##            normal(3,1)     b Intercept                       c <NA> <NA>
##          normal(0,0.5)     b                                 a <NA> <NA>
##            normal(2,1)     b Intercept                       a <NA> <NA>
##            constant(0)     b Intercept                       b <NA> <NA>
##        source
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
##  (vectorized)
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##       default
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##  (vectorized)
##  (vectorized)
##          user
```

```
##          user
##          user
##          user
##          user
```

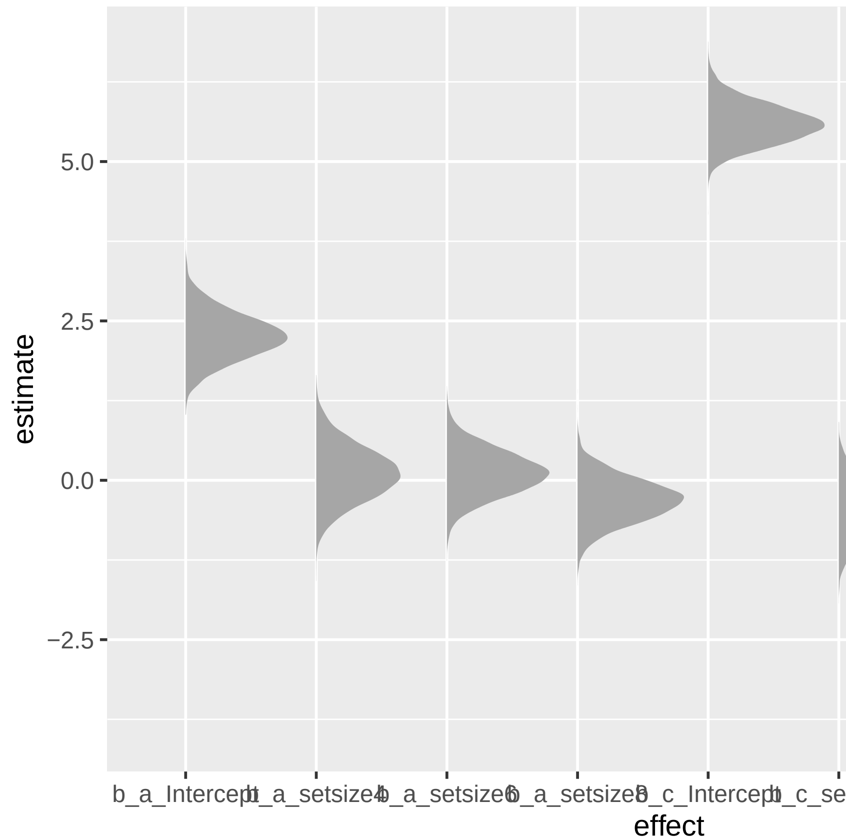
### Fit and run model

```
##   Model: m3(resp_cats = c("IIP", "IOP", "NPL"),
##             num_options = c("n_IIP", "n_IOP", "n_NPL"),
##             choice_rule = "softmax",
##             version = "ss")
##   Links: c = identity; a = identity
## Formula: b = 0
##          c ~ 1 + setsize + (1 + setsize | id)
##          a ~ 1 + setsize + (1 + setsize | id)
##          IIP ~ b + a + c
##          IOP ~ b + a
##          NPL ~ b
##    Data: (Number of observations: 480)
##   Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##          total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 20)
##                              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## sd(c_Intercept)                  0.39      0.16     0.05     0.71 1.01      931
## sd(c_setsize4)                   0.51      0.28     0.04     1.12 1.00     1777
## sd(c_setsize6)                   0.44      0.19     0.12     0.86 1.00      867
## sd(c_setsize8)                   0.22      0.15     0.01     0.56 1.01      615
## sd(a_Intercept)                  0.67      0.33     0.06     1.36 1.00     1546
## sd(a_setsize4)                   0.75      0.58     0.03     2.18 1.00     3840
## sd(a_setsize6)                   0.47      0.36     0.02     1.36 1.00     2272
## sd(a_setsize8)                   0.46      0.33     0.02     1.21 1.00     1431
## cor(c_Intercept,c_setsize4)      0.02      0.40    -0.72     0.78 1.00     3771
## cor(c_Intercept,c_setsize6)      0.37      0.37    -0.50     0.91 1.00     1020
## cor(c_setsize4,c_setsize6)       0.18      0.39    -0.64     0.82 1.00     1691
## cor(c_Intercept,c_setsize8)     -0.11      0.44    -0.84     0.74 1.00     3304
## cor(c_setsize4,c_setsize8)       0.04      0.44    -0.77     0.81 1.00     2180
## cor(c_setsize6,c_setsize8)       0.33      0.46    -0.66     0.94 1.00      875
## cor(a_Intercept,a_setsize4)     -0.04      0.44    -0.83     0.79 1.00    10775
## cor(a_Intercept,a_setsize6)     -0.01      0.44    -0.80     0.81 1.00     6422
## cor(a_setsize4,a_setsize6)       0.02      0.45    -0.82     0.83 1.00     6460
## cor(a_Intercept,a_setsize8)     -0.01      0.44    -0.79     0.81 1.00     5913
## cor(a_setsize4,a_setsize8)       0.04      0.45    -0.80     0.82 1.00     3685
## cor(a_setsize6,a_setsize8)       0.15      0.45    -0.76     0.88 1.00     2572
##                              Tail_ESS
## sd(c_Intercept)                   923
## sd(c_setsize4)                   2544
## sd(c_setsize6)                   1339
## sd(c_setsize8)                   1509
## sd(a_Intercept)                  1434
## sd(a_setsize4)                   3108
## sd(a_setsize6)                   3879
## sd(a_setsize8)                   3287
```

```
## cor(c_Intercept,c_setsize4)     3714
## cor(c_Intercept,c_setsize6)      773
## cor(c_setsize4,c_setsize6)      4106
## cor(c_Intercept,c_setsize8)     3489
## cor(c_setsize4,c_setsize8)      4984
## cor(c_setsize6,c_setsize8)      1883
## cor(a_Intercept,a_setsize4)     5116
## cor(a_Intercept,a_setsize6)     5452
## cor(a_setsize4,a_setsize6)      5636
## cor(a_Intercept,a_setsize8)     5404
## cor(a_setsize4,a_setsize8)      5282
## cor(a_setsize6,a_setsize8)      5527
##
## Regression Coefficients:
##             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## c_Intercept     5.59      0.31     5.02     6.22 1.00     3593     4229
## c_setsize4     -0.46      0.37    -1.18     0.26 1.00     3787     5553
## c_setsize6     -2.03      0.32    -2.70    -1.43 1.00     3425     4598
## c_setsize8     -2.86      0.31    -3.49    -2.29 1.00     3504     4124
## a_Intercept     2.26      0.36     1.55     2.98 1.00     5165     5497
## a_setsize4      0.10      0.42    -0.71     0.96 1.00    10348     6081
## a_setsize6      0.11      0.35    -0.56     0.81 1.00     6661     6472
## a_setsize8     -0.31      0.34    -0.99     0.35 1.00     5753     5885
##
## Constant Parameters:
##                 Value
## b_Intercept      0.00
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

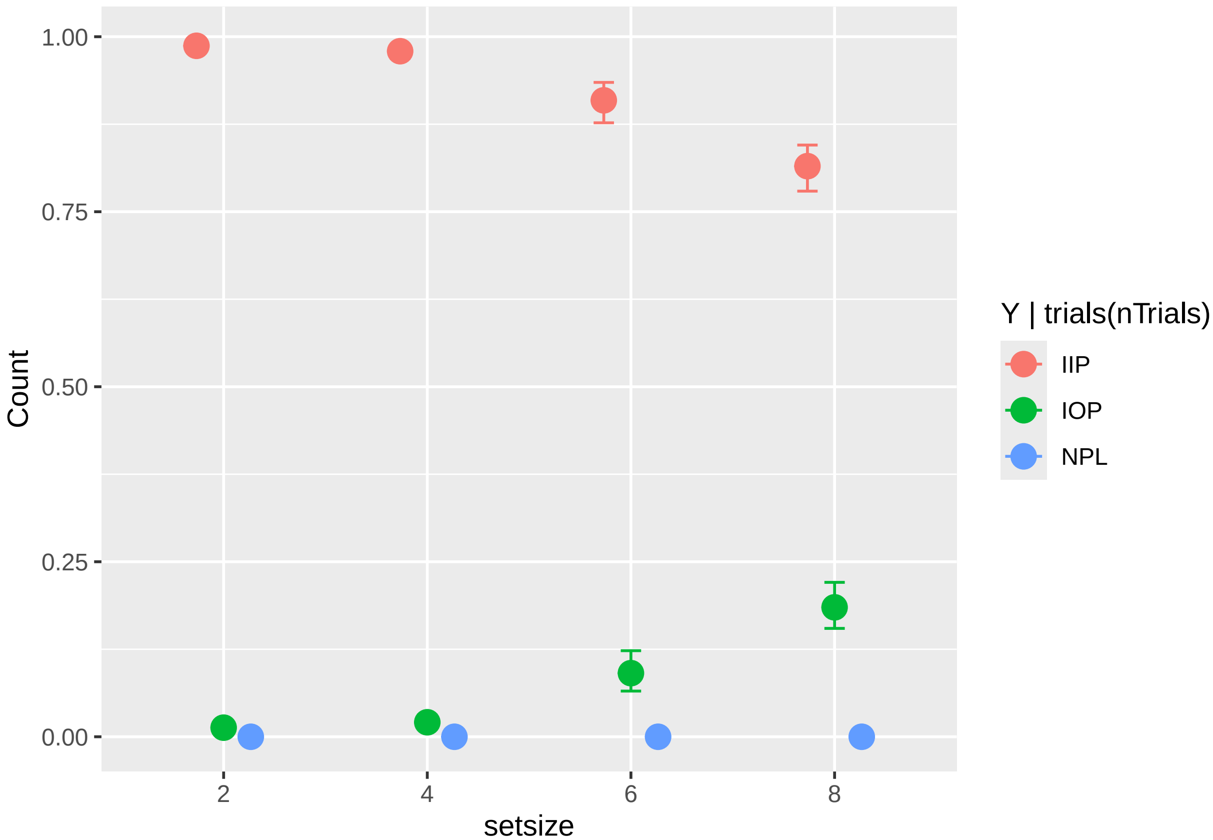
```r
# extract posterior samples
posterior <- as_draws_df(m3_fit_ss_exp1) %>%
  select(starts_with("b_")) %>%
  pivot_longer(cols = starts_with("b_"), names_to = "effect", values_to = "estimate") %>%
  filter(effect != "b_b_Intercept")

# plot posterior samples
ggplot(posterior, aes(x = effect, y = estimate)) +
  stat_slab()
```




**Plot model posteriors and model predicitions**

```
# plot model predictions on native scale
conditional_effects(m3_fit_ss_exp1, categorical = T, effects = "setsize")
```




```r
# to-be testes hypotheses
tests <- c(
  "c_Intercept = c_setsize4",
  "c_setsize4 = c_setsize6",
  "c_setsize6 = c_setsize8",
  "a_Intercept = a_setsize4",
  "a_setsize4 = a_setsize6",
  "a_setsize6 = a_setsize8"
)

hypothesis(m3_fit_ss_exp1, tests)
```

### Hypothesis tests

```
## Hypothesis Tests for class b:
##                 Hypothesis Estimate Est.Error CI.Lower CI.Upper Evid.Ratio
## 1 (c_Intercept)-(c_... = 0     6.06      0.63     4.86     7.35       0.00
## 2 (c_setsize4)-(c_s... = 0     1.57      0.25     1.11     2.09       0.00
## 3 (c_setsize6)-(c_s... = 0     0.82      0.12     0.59     1.07       0.00
## 4 (a_Intercept)-(a_... = 0     2.15      0.62     0.96     3.36       0.01
## 5 (a_setsize4)-(a_s... = 0    -0.01      0.51    -0.98     1.01       1.40
## 6 (a_setsize6)-(a_s... = 0     0.43      0.33    -0.21     1.08       0.93
##   Post.Prob Star
## 1      0.00    *
## 2      0.00    *
## 3      0.00    *
```

```
## 4      0.01    *
## 5      0.58
## 6      0.48
## ---
## 'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.
## '*': For one-sided hypotheses, the posterior probability exceeds 95%;
## for two-sided hypotheses, the value tested against lies outside the 95%-CI.
## Posterior probabilities of point hypotheses assume equal prior probabilities.
```

## Task 2: Experiment 2

In experiment 2, the same design as in experiment 1 was applied. However, the pool from which the list words were chosen from was restricted in comparison to experiment 1. We want to examine whether and how this switch from a large word pool in experiment 1 to a small word pool in experiment 2 changes the underlying cognitive processes of the set size effect.

a) Thus, we will first plot the data from experiment 2 descriptively ("exercise3_data_exp2.txt").
b) Next, we implement the same basic M3 model and examine the model output. Plot the model results.
c) We test whether there are differences among the different set sizes using hypothesis tests. Do the results differ between experiment 1 and 2? If yes, how so?

### Load data

```r
# experiment 2
mydata_exp2 <- read.table(here('data', 'exercise3_data_exp2.txt'))
mydata_exp2$setsize <- as.factor(mydata_exp2$setsize)
```

### Descriptive proportions

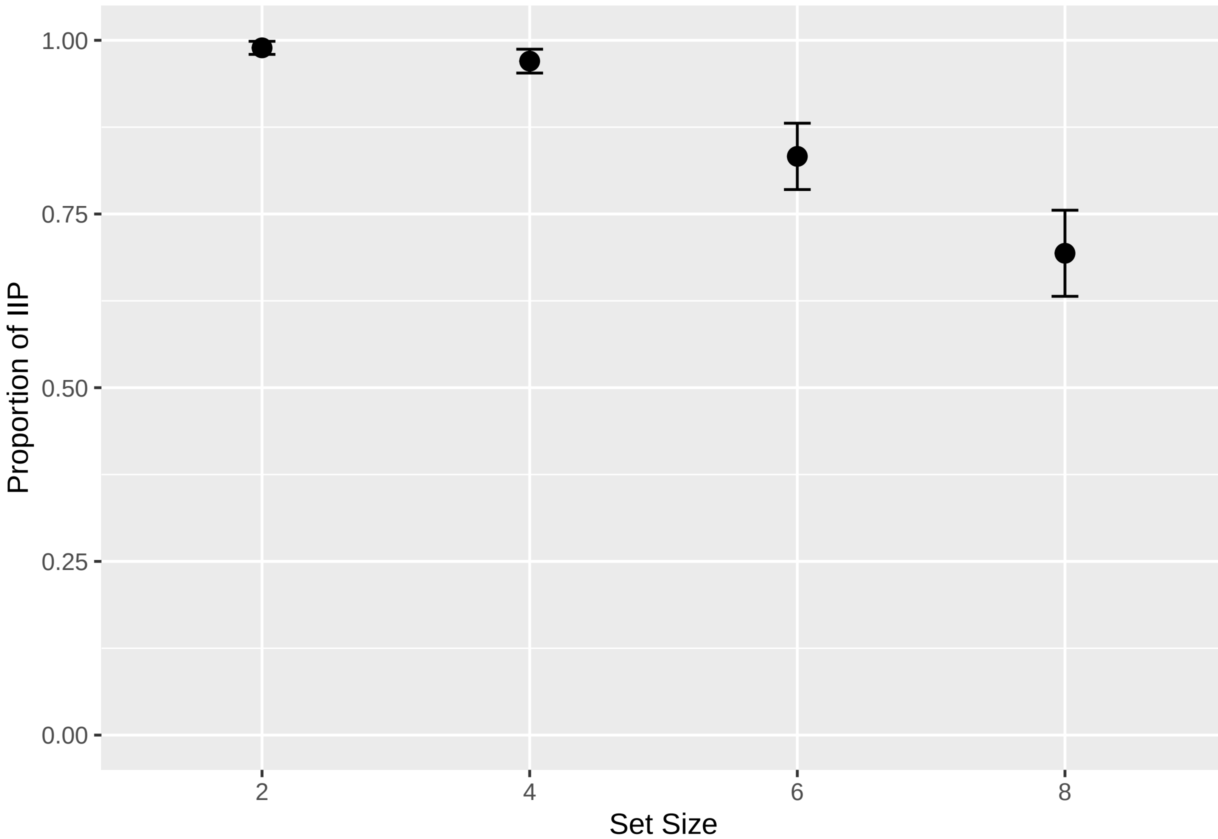
```r
# calculate proportions
prop_data <- mydata_exp2 %>%
  group_by(id, setsize) %>%
  summarise(prop = sum(IIP) / sum(total), .groups = "drop") %>%
  group_by(setsize) %>%
  summarise(
    mean_prop = mean(prop),
    se = sd(prop) / sqrt(n()),
    lower = mean_prop - 1.96 * se,
    upper = mean_prop + 1.96 * se,
    .groups = "drop"
  )

prop_data
```

```
## # A tibble: 4 x 5
##   setsize mean_prop      se lower upper
##   <fct>       <dbl>   <dbl> <dbl> <dbl>
## 1 2           0.989 0.00480 0.980 0.999
## 2 4           0.97  0.00875 0.953 0.987
## 3 6           0.833 0.0244  0.785 0.881
## 4 8           0.693 0.0316  0.632 0.755
```

```r
# plot proportions
ggplot(prop_data, aes(x = factor(setsize), y = mean_prop)) +
  geom_point(size = 3) +
```

```r
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.1) +
  labs(y = "Proportion of IIP", x = "Set Size") +
  coord_cartesian(ylim = c(0, 1))
```




**Implement basic M3 model**

```r
n_cores <- 4
n_iters <- 2000 # number of total iterations
n_warmup <- 1000 # number of warmups
n_chains <- 4 # number of chains
```

**Set model settings**

```r
# model
m3_model_ss <- m3(resp_cats = c("IIP", "IOP", "NPL"),
               num_options = c("n_IIP", "n_IOP", "n_NPL"),
               choice_rule = "softmax",
               version = "ss")

# model formula
m3_formula_ss <- bmf(
  IIP ~ b + a + c,
  IOP ~ b + a,
  NPL ~ b,
  c ~ 1 + setsize + (1 + setsize | id),
```

```r
  a ~ 1 + setsize + (1 + setsize | id)
)

# check priors
default_prior(m3_formula_ss, data = mydata_exp2, model = m3_model_ss)
```

**Define model formula, priors and model**

```
##                  prior class      coef group resp dpar nlpar   lb   ub
##                 lkj(1)   cor
##                 lkj(1)   cor               id
##         normal(0,0.5)     b  setsize4                     a <NA> <NA>
##         normal(0,0.5)     b  setsize6                     a <NA> <NA>
##         normal(0,0.5)     b  setsize8                     a <NA> <NA>
##  student_t(3, 0, 2.5)    sd                               a    0
##  student_t(3, 0, 2.5)    sd               id              a    0
##  student_t(3, 0, 2.5)    sd Intercept     id              a    0
##  student_t(3, 0, 2.5)    sd  setsize4     id              a    0
##  student_t(3, 0, 2.5)    sd  setsize6     id              a    0
##  student_t(3, 0, 2.5)    sd  setsize8     id              a    0
##                 (flat)     b                              b
##           normal(0,2)     b  setsize4                     c <NA> <NA>
##           normal(0,2)     b  setsize6                     c <NA> <NA>
##           normal(0,2)     b  setsize8                     c <NA> <NA>
##  student_t(3, 0, 2.5)    sd                               c    0
##  student_t(3, 0, 2.5)    sd               id              c    0
##  student_t(3, 0, 2.5)    sd Intercept     id              c    0
##  student_t(3, 0, 2.5)    sd  setsize4     id              c    0
##  student_t(3, 0, 2.5)    sd  setsize6     id              c    0
##  student_t(3, 0, 2.5)    sd  setsize8     id              c    0
##           normal(0,2)     b                               c <NA> <NA>
##           normal(3,1)     b Intercept                     c <NA> <NA>
##         normal(0,0.5)     b                               a <NA> <NA>
##           normal(2,1)     b Intercept                     a <NA> <NA>
##           constant(0)     b Intercept                     b <NA> <NA>
##        source
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
##  (vectorized)
##  (vectorized)
```

```
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##          user
##          user
##          user
##          user
```

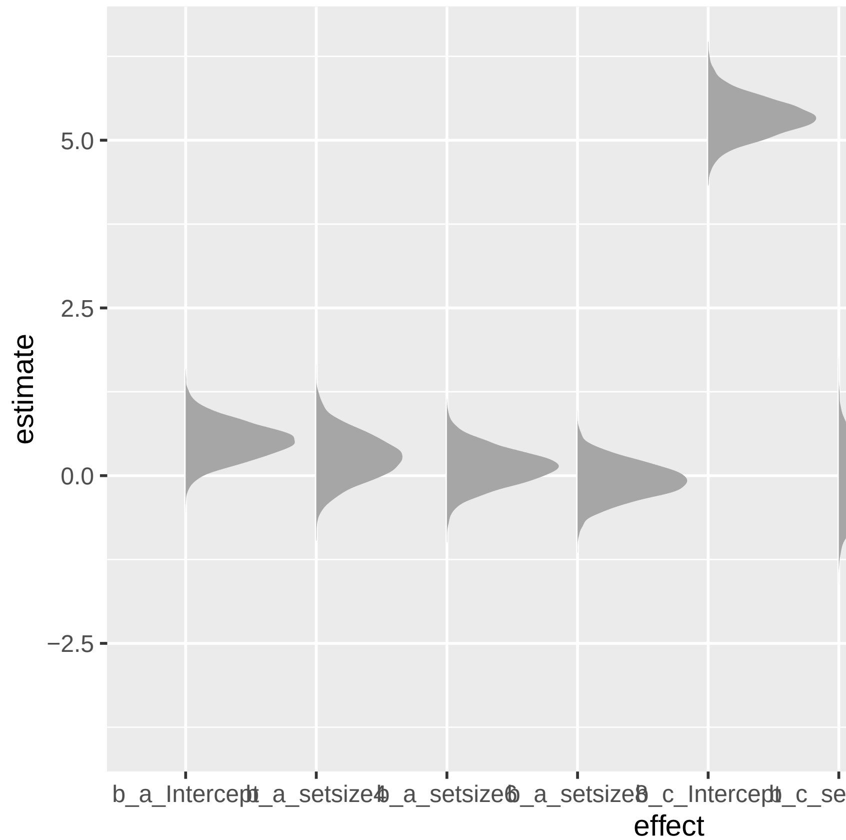
### Fit and run model

```
##   Model: m3(resp_cats = c("IIP", "IOP", "NPL"),
##             num_options = c("n_IIP", "n_IOP", "n_NPL"),
##             choice_rule = "softmax",
##             version = "ss")
##   Links: c = identity; a = identity
## Formula: b = 0
##          c ~ 1 + setsize + (1 + setsize | id)
##          a ~ 1 + setsize + (1 + setsize | id)
##          IIP ~ b + a + c
##          IOP ~ b + a
##          NPL ~ b
##    Data: (Number of observations: 480)
##   Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##          total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~id (Number of levels: 20)
##                              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## sd(c_Intercept)                  0.80      0.19     0.47     1.22 1.00     2331
## sd(c_setsize4)                   0.94      0.39     0.27     1.80 1.00     1340
## sd(c_setsize6)                   0.40      0.21     0.06     0.90 1.01      533
## sd(c_setsize8)                   0.24      0.19     0.01     0.71 1.00      552
## sd(a_Intercept)                  0.31      0.14     0.07     0.62 1.00     2933
## sd(a_setsize4)                   0.49      0.39     0.02     1.43 1.00     3577
## sd(a_setsize6)                   0.25      0.18     0.01     0.67 1.00     2796
## sd(a_setsize8)                   0.16      0.13     0.01     0.48 1.00     2787
## cor(c_Intercept,c_setsize4)      0.14      0.30    -0.43     0.75 1.00     2468
## cor(c_Intercept,c_setsize6)      0.03      0.34    -0.61     0.70 1.00     1705
## cor(c_setsize4,c_setsize6)      -0.12      0.35    -0.77     0.57 1.00     1892
## cor(c_Intercept,c_setsize8)     -0.07      0.42    -0.77     0.76 1.00     3075
## cor(c_setsize4,c_setsize8)      -0.03      0.43    -0.81     0.77 1.00     3221
## cor(c_setsize6,c_setsize8)       0.41      0.45    -0.66     0.96 1.01      796
## cor(a_Intercept,a_setsize4)     -0.00      0.43    -0.78     0.79 1.00     7863
## cor(a_Intercept,a_setsize6)     -0.02      0.43    -0.79     0.78 1.00     6344
## cor(a_setsize4,a_setsize6)      -0.03      0.44    -0.81     0.79 1.00     5260
## cor(a_Intercept,a_setsize8)     -0.21      0.46    -0.90     0.72 1.00     6717
## cor(a_setsize4,a_setsize8)       0.06      0.45    -0.79     0.83 1.00     5695
## cor(a_setsize6,a_setsize8)       0.09      0.45    -0.79     0.85 1.00     5413
##                              Tail_ESS
## sd(c_Intercept)                  3090
## sd(c_setsize4)                   1085
## sd(c_setsize6)                    977
## sd(c_setsize8)                    782
```

```
## sd(a_Intercept)                2870
## sd(a_setsize4)                 3572
## sd(a_setsize6)                 4246
## sd(a_setsize8)                 4208
## cor(c_Intercept,c_setsize4)    2583
## cor(c_Intercept,c_setsize6)    2731
## cor(c_setsize4,c_setsize6)     3277
## cor(c_Intercept,c_setsize8)    4837
## cor(c_setsize4,c_setsize8)     4424
## cor(c_setsize6,c_setsize8)     1973
## cor(a_Intercept,a_setsize4)    5657
## cor(a_Intercept,a_setsize6)    5727
## cor(a_setsize4,a_setsize6)     5561
## cor(a_Intercept,a_setsize8)    5034
## cor(a_setsize4,a_setsize8)     5767
## cor(a_setsize6,a_setsize8)     6548
##
## Regression Coefficients:
##             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## c_Intercept     5.34      0.29     4.78     5.91 1.00     2829     3829
## c_setsize4     -0.11      0.39    -0.83     0.72 1.00     3128     4775
## c_setsize6     -2.17      0.27    -2.72    -1.65 1.00     3365     4078
## c_setsize8     -2.87      0.26    -3.40    -2.40 1.00     3391     4440
## a_Intercept     0.52      0.27    -0.00     1.05 1.00     3164     4442
## a_setsize4      0.28      0.34    -0.37     0.94 1.00     5926     5435
## a_setsize6      0.13      0.27    -0.41     0.66 1.00     3279     4632
## a_setsize8     -0.07      0.26    -0.58     0.43 1.00     3184     4484
##
## Constant Parameters:
##                 Value
## b_Intercept      0.00
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

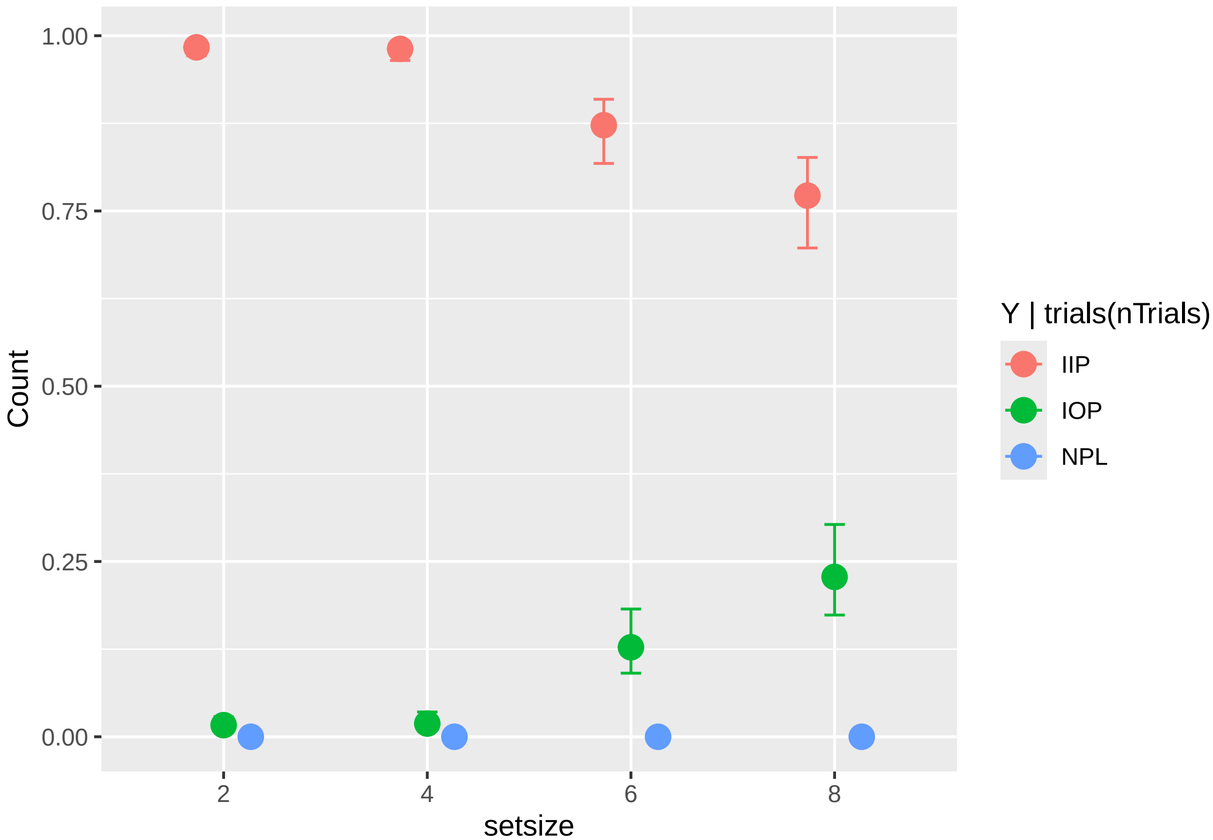
```r
# extract posterior samples
posterior <- as_draws_df(m3_fit_ss_exp2) %>%
  select(starts_with("b_")) %>%
  pivot_longer(cols = starts_with("b_"), names_to = "effect", values_to = "estimate") %>%
  filter(effect != "b_b_Intercept")

# plot posterior samples
ggplot(posterior, aes(x = effect, y = estimate)) +
  stat_slab()
```




**Plot model posteriors and model predicitions**

```
# plot model predictions on native scale
conditional_effects(m3_fit_ss_exp2, categorical = T, effects = "setsize")
```




```r
# to-be testes hypotheses
tests <- c(
  "c_Intercept = c_setsize4",
  "c_setsize4 = c_setsize6",
  "c_setsize6 = c_setsize8",
  "a_Intercept = a_setsize4",
  "a_setsize4 = a_setsize6",
  "a_setsize6 = a_setsize8"
)

hypothesis(m3_fit_ss_exp2, tests)
```

**Hypothesis tests**

```
## Hypothesis Tests for class b:
##                 Hypothesis Estimate Est.Error CI.Lower CI.Upper Evid.Ratio
## 1 (c_Intercept)-(c_... = 0     5.45      0.58     4.29     6.59       0.00
## 2 (c_setsize4)-(c_s... = 0     2.06      0.34     1.46     2.81       0.00
## 3 (c_setsize6)-(c_s... = 0     0.70      0.10     0.50     0.91       0.00
## 4 (a_Intercept)-(a_... = 0     0.24      0.52    -0.77     1.25       9.29
## 5 (a_setsize4)-(a_s... = 0     0.15      0.34    -0.48     0.84       2.02
## 6 (a_setsize6)-(a_s... = 0     0.20      0.15    -0.10     0.51       1.95
##   Post.Prob Star
## 1      0.00    *
## 2      0.00    *
## 3      0.00    *
```

```
## 4      0.90
## 5      0.67
## 6      0.66
## ---
## 'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.
## '*': For one-sided hypotheses, the posterior probability exceeds 95%;
## for two-sided hypotheses, the value tested against lies outside the 95%-CI.
## Posterior probabilities of point hypotheses assume equal prior probabilities.
```